

Senal Liyanage, Ph.D.

Computational Chemist | Molecular Modeling | Molecular Dynamics | HPC Scientific Computing

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PROFESSIONAL SUMMARY

Computational chemist with Ph.D. training in molecular simulation, quantum chemistry, protein-ligand modeling, polymer/nanoparticle systems, and high-performance scientific computing. Experienced in building reproducible MD, docking-oriented, QM, and trajectory-analysis workflows with Python/Bash on Linux, Slurm, and GPU-enabled research clusters. Current work focuses on PLGA nanoparticles, ionic-liquid interfacial reorganization, drug-delivery modeling, scientific visualization, and public-facing research software/portfolio development.

CORE COMPETENCIES

Molecular simulation	GROMACS, AMBER, NAMD, OpenMM; MD setup, equilibration, production, replicate analysis, trajectory post-processing
Modeling and quantum chemistry	Protein-ligand modeling, docking-oriented workflows, DFT/TDDFT, QM/MM, Q-Chem, Gaussian, ORCA, GAMESS
Programming and analysis	Python, Bash, FORTRAN, NumPy, pandas, scikit-learn, Matplotlib, MDAnalysis, GraphPad Prism
HPC and workflow support	Linux, Slurm, GPU job workflows, conda/module environments, multi-user research computing, reproducible documentation

PROFESSIONAL EXPERIENCE

Postdoctoral Fellow | Department of Chemistry and Physics, Tougaloo College | Jan 2026-Present

- Lead atomistic molecular simulation study of PLGA nanoparticles in water and ionic-liquid/water environments; evaluate polymer mass envelope, hydration displacement, ion localization, and interfacial orientation across replicate trajectories.
- Develop manuscript-level analysis and visualization workflows using GROMACS-compatible trajectories, Python/Bash, MDAnalysis-style scripting, VMD, ChimeraX, and Matplotlib.
- Support Linux/Slurm GPU research-computing workflows, including user accounts, conda/module environments, GROMACS/OpenMM/Schrodinger jobs, and troubleshooting of scheduler/GPU issues.
- Mentor students and collaborators in docking, MD setup, trajectory analysis, and reproducible computational chemistry workflows.

Doctoral Researcher | Department of Chemistry, Mississippi State University | Jan 2020-Dec 2025

- Completed Ph.D. dissertation, "Computational modeling of environment-driven molecular interactions in biological systems"; advisor: Dr. Steven R. Gwaltney.
- Performed molecular dynamics, QM/MM, free-energy, and quantum-chemical modeling of biomolecular recognition, membrane remodeling, solvatochromic behavior, and protein-ligand interactions.
- Built Python/Bash/HPC workflows to process MD trajectories and electronic-structure outputs; generated publication-ready figures, summaries, and mechanistic interpretations.
- Applied AMBER, NAMD, GROMACS, Q-Chem, GAMESS, Gaussian, VMD, PyMOL, ChimeraX, NumPy, pandas, and Matplotlib across research projects.

Researcher | Institute of Chemistry Ceylon, Sri Lanka | Oct 2017-Dec 2019

- Conducted chemistry research and analysis during M.Phil. training, supporting molecular modeling, literature-driven project design, and scientific communication.

Visiting Lecturer | Rajarata University of Sri Lanka | Dec 2025-Present

- Teach and support chemistry learning activities with emphasis on clear scientific explanation, quantitative reasoning, and student development.

SELECTED PORTFOLIO PROJECTS

chemistry-analysis-tools - Public GitHub repository with chemistry utilities and MD-analysis scaffold scripts for radial density, radius-shell counts, and orientation metrics.

polymer-tool - PLGA-first model-preparation workflow for chain construction, nanoparticle seed packing, packing assessment, provenance, and downstream handoff bundles.

qctddft - Command-line toolkit for Q-Chem TDDFT post-processing, broadened spectra, spectral-region assignment, and representative-structure clustering.

SELECTED TECHNICAL EVIDENCE

PLGA/IL nanoparticle simulations	Designed and analyzed atomistic MD workflows for polymer mass-envelope metrics, water displacement, ion localization, and interfacial orientation.
Research computing support	Configured and troubleshot Slurm/GPU workflows, scientific Python environments, GROMACS/OpenMM execution, and multi-user cluster access.
Scientific communication	Prepared manuscript-oriented figures, graphical abstracts, cover letters, poster materials, and research explanations for computational chemistry audiences.
Public software practice	Maintains GitHub repositories with READMEs, reproducibility notes, data-policy boundaries, and portfolio-ready scientific code structure.

EDUCATION

Ph.D., Computational Chemistry | Mississippi State University | Jan 2020-Dec 2025 | Dissertation: Computational modeling of environment-driven molecular interactions in biological systems

M.Phil., Chemistry | University of Colombo, Sri Lanka | Jan 2018-Dec 2019

B.Sc., Chemistry | Institute of Chemistry Ceylon, Sri Lanka | Jan 2014-Dec 2017

SELECTED PUBLICATIONS AND PRESENTATIONS

- "Computational Modeling to Understand the Interaction of TMPyP4 with a G-Quadruplex," Journal of Biomolecular Structure and Dynamics, 2025.
- "An All-Atom Model of the Human Cardiac Sodium Channel in a Lipid Bilayer," Scientific Reports, 2024.
- Accepted poster: "Multiscale molecular simulations of ionic-liquid-modified PLGA nanoparticles for controlled medicine load and release," ACS Fall 2026, MEDI Division.

TARGET ROLES

Computational chemistry / molecular modeling	Computational Chemist, Molecular Modeling Scientist, Postdoctoral Scientist, Modeling Scientist, Drug Delivery Modeling Scientist
Scientific computing / HPC	Scientific Computing Specialist, Research Computing Specialist, HPC Applications Scientist, Computational Research Scientist